

Zhuoyun (Joey) Zhong

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PROFILE

Zhuoyun is a Ph.D. candidate in Robotics Engineering at Worcester Polytechnic Institute (WPI), specializing in **Sampling-based Motion Planning, Humanoid Whole-body Planning, Active Robot Learning, and Non-prehensile Manipulation**. Recipient of the **ICRA Best Student Paper Award**, with first-author publications at ICRA and IROS. Experienced in developing learning- and planning-based algorithms and frameworks, with hands-on experience in real-world robot deployment and system integration.

EDUCATION

Worcester Polytechnic Institute (WPI)

Ph.D. in Robotics Engineering, GPA: 4.00/4.00

Aug. 2020 – Present

Jilin University (JLU), China

B.S. in Automotive Engineering, GPA: 3.49/4.00

Sep. 2015 – Jul. 2019

SKILLS

System and Software - OMPL, MuJoCo, PyTorch, ROS2, Unity

Programming - Python, C++, C#, MATLAB

Language - English, Chinese Mandarin, Cantonese

WORK EXPERIENCE

Siemens Corporation – Advanced Robotics and AI Intern, Berkeley, CA

Jul. 2026 – Sep. 2026

- Addressed autonomous building inspection with humanoid robots and foundation models, where manual facility inspections are labor-intensive and difficult to scale.
- Developed a task-adaptive humanoid inspection framework that integrates LLM-based Behavior Tree synthesis, a library of reusable robot skills, Behavior Tree execution, and VLM-based visual inspection to adapt inspection workflows to different targets and requirements.
- Deployed the complete system on a Unitree G1 humanoid robot and evaluated real-world inspection tasks in an operational building environment.

RESEARCH EXPERIENCE

Hierarchical Humanoid Motion Planning From a Single Human Demonstration

Sep. 2025 – Present

- Developed a humanoid planning algorithm to learn from a single human video, where demonstrated motions must be adapted to new start, goal, and obstacle arrangements while preserving task-specific motion behavior.
- Proposed Phase-RRT*, a demonstration-guided sampling-based planner that enforces collision avoidance while minimizing deviation from the re-targeted motion reference and preserving demonstration progress.
- Integrated global planning, optimization-based local trajectory correction, and a learned whole-body RL controller for accurate motion execution in MuJoCo and on the physical Unitree G1.

KiT: Kinodynamic Planning with a Terminal Cost and Learned Uncertainty

May 2025 – May 2026

- Addressed kinodynamic planning under uncertainty, where existing planners typically optimize only running cost and do not explicitly account for terminal-state desirability or goal-reaching probability.
- Proposed kinodynamic planning with terminal cost and proved that AO-RRT remains asymptotically optimal.
- Derived a terminal objective with Wasserstein distance that lower-bounds goal-reaching probability.
- Learned system dynamics and uncertainty directly from data, and integrated them into the planners.
- Validated in MuJoCo simulation and real-world tasks including Flappy Bird, Navigation, and Planar Pushing, demonstrating explicit terminal-state optimization while achieving significantly higher success rate.

COAD: Constant-Time Planning with Compressed Library and Online Adaptation

Nov. 2025 – Mar. 2026

- Tackled repeated manipulation planning in continuous goal-varying settings.
- Developed a framework for constant-time planning via task-space discretization and a compressed motion library with lightweight online adaptation, such as interpolation, DMPs, and optimization.
- Evaluated in simulation and real-world manipulation tasks, achieving up to 98% library compression ratio and sub-millisecond planning time.

ActivePusher: Active Learning and Planning for Nonprehensile Manipulation Nov. 2024 – Sep. 2025

- Developed a framework for nonprehensile manipulation with a learned dynamics model under uncertainty, where data collection is expensive and contact-rich pushing dynamics are hard to predict accurately.
- Used NTK-based epistemic uncertainty estimation and the BAIT strategy for data-efficient model learning.
- Integrated model uncertainty into motion planning to bias action selection toward more reliable regions.
- Validated the method in both MuJoCo simulation and real-world pushing tasks, showing improved data efficiency, higher task success, and lower execution error compared with baseline methods.

Expansion-GRR: Efficient Generation of Smooth Global Redundancy Roadmaps Nov. 2023 – Apr. 2024

- Addressed global redundancy resolution (GRR) for redundant manipulators, where prior GRR methods often required long computation times and produced less smooth task-constrained motions.
- Developed an efficient GRR generation method that leverages configuration-space projection and expansion.
- Proposed a multi-seed strategy and continuity-aware neighbor expansion to effectively generate roadmaps.
- Built a teleoperation pipeline using Expansion-GRR roadmaps that outperformed prior methods and reactive IK solvers in success rate and solution quality, with roadmap generation up to two orders of magnitude faster.

NurSim: A Simulation for Intelligent Robotic Nursing Assistant Development May 2021 – Jan. 2024

- Developed a Unity-based simulation environment for nursing robots and human-robot interaction studies.
- Built ROS-compatible digital twins of nursing robots with GUI and VR/XR teleoperation interfaces.
- Evaluated the platform through pilot studies as a testbed for interface design and HRI in nursing tasks.

Self-supervised Pre-training for Generic Spatial-Temporal Representations Jan. 2023 – Jul. 2023

- Developed a Transformer-based self-supervised pre-training framework for spatial-temporal representations.
- Designed a sub-trajectory similarity objective and fine-tuning pipeline for downstream tasks.
- Improved downstream accuracy by 16.2% and 13.0% over average baselines on real vehicle trajectory dataset.

PUBLICATIONS

Under Review

- **Zhuoyun Zhong**, Seyedali Golestaneh and Constantinos Chamzas, “Terminal Matters: Kinodynamic Planning in Belief State-Cost Space with a Terminal Cost and Learned Uncertainty”, Submitted to IEEE Transactions on Robotics (T-RO).
- Adil Shiyas*, **Zhuoyun Zhong*** and Constantinos Chamzas, “CoAd: Constant-Time Planning for Continuous Goal Manipulation with Compressed Library and Online Adaptation”, Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027.

Accepted / Published

- Seyedali Golestaneh, **Zhuoyun Zhong** and Constantinos Chamzas, “AURA: Asymptotically Optimal Uncertainty Robust Replanning Algorithm for Kinodynamic Systems”, Accepted by IEEE Robotics and Automation Letters (RA-L), 2026.
- **Zhuoyun Zhong**, Seyedali Golestaneh and Constantinos Chamzas, “ActivePusher: Active Learning and Planning with Residual Physics for Nonprehensile Manipulation”, Accepted by IEEE International Conference on Robotics and Automation (ICRA), 2026. **ICRA Best Student Paper Award, Best Paper Award on Planning and Control Finalists.**
- **Zhuoyun Zhong**, Zhi Li and Constantinos Chamzas, “Expansion-GRR: Efficient Generation of Smooth Global Redundancy Resolution Roadmaps”, Accepted by IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024.
- Nikita Boguslavskii*, **Zhuoyun Zhong***, Lorena Maria Genua* and Zhi Li, “A Shared Autonomous Nursing Robot Assistant with Dynamic Workspace for Versatile Mobile Manipulation”, Accepted by IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023.
- Mingzhi Hu, **Zhuoyun Zhong**, Xin Zhang, Yanhua Li, Yiqun Xie, Xiaowei Jia, Xun Zhou, and Jun Luo, “Self-supervised pre-training for robust and generic spatial-temporal representations”, Accepted by IEEE International Conference on Data Mining (ICDM), 2023.